higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE

DIGITAL ELECTRONICS N4

(8080354)

12 April 2024 (X-paper)
09:00–12:00

Drawing instruments may be used. Calculators may NOT be used.

This question paper consists of 7 pages.

247Q1A2412

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DEPARTMENT OF HIGHER EDUCATION AND TRAINING
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE
DIGITAL ELECTRONICS N4
TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer all the questions.
2. Read all the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Start each section on a new page.
5. Use only a black or blue pen.
6. Write neatly and legibly.

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SECTION A

QUESTION 1

Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1–1.5) in the ANSWER BOOK.

1.1 $3,3_{16} + 3,3_{8}$ =

A 6,6
B 6,44
C $6,44_{16}$
D $6,44_{8}$

1.2 $10,1_{8} + 10,125$ =

A 18,5
B 18,25
C 18,125
D $10010,11_{2}$

1.3 $10010,11_{2} – 10,625$ =

A $8,2_{16}$
B $8,8_{8}$
C 8,8
D $100,01_{2}$

1.4 $1,1_{2} \times 2,8_{16}$ =

A $3,C_{16}$
B $11,1101_{2}$
C $3,D_{16}$
D $10,1101_{2}$

1.5 $3,6_{8} \div 1,8_{16}$ =

A $10,1_{8}$
B $10,1_{16}$
C 2,25
D $2,8_{16}$

(5 × 2) **[10]**

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QUESTION 2

Indicate whether the following statements are TRUE or FALSE by writing only 'True' or 'False' next to the question number (2.1–2.8) in the ANSWER BOOK.

2.1 $\overline{A} + AB = \overline{A} + B$

2.2 $\overline{A} + AB = \overline{\overline{A} + B}$

2.3 $\overline{X} + YZ + 1 = 1$

2.4 $A + B + \overline{A} = 1$

2.5 $X + YZ + 0 = X + YZ$

2.6 $\overline{\overline{A} + B\overline{B}} = \overline{A} \oplus B$

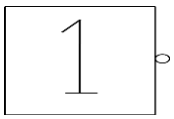
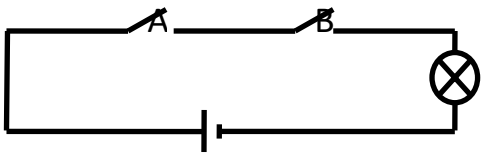
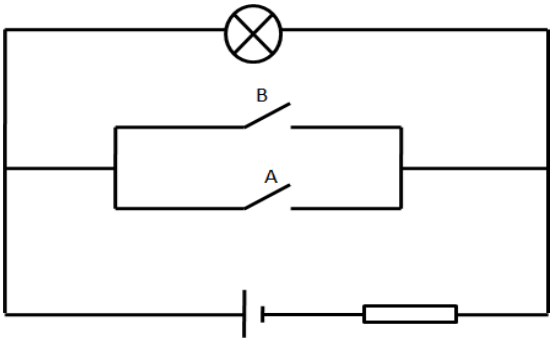
2.7 $A\overline{B} + \overline{AB} = A \oplus B$

2.8 $\overline{A}B + A\overline{B} = A \oplus B$

(8 × 1) **[8]**

QUESTION 3

Choose a term from COLUMN B that matches a description in COLUMN A. Write only the letter (A–G) next to the question number (3.1–3.7) in the ANSWER BOOK.

COLUMN A		COLUMN B
3.1	$A\overline{B} + \overline{A}B = O/P$	A OR
3.2	$\overline{A \oplus B} = O/P$	B NOR
3.3		C AND
		D NAND
		E EXCLUSIVE OR
3.4	A B O/P 0 0 1 0 1 1 1 0 1 1 1 0	F EXCLUSIVE NOR
		G NOT
3.5	A B O/P 0 0 0 0 1 1 1 0 1 1 1 1	
3.6		
3.7		

(7 × 1) **[7]**

TOTAL SECTION A: 25

SECTION B

QUESTION 4

Complete the following calculations in the binary numbering system. All steps must be indicated clearly.

4.1 $-1010{,}101_2 - 1100{,}1_2$ (Make use of the 1's complement method) (5)

4.2 $1101{,}01111_2 - 1110{,}1001_2$ (Make use of the 2's complement method) (5)

4.3 $10100{,}1101_2 \times 101{,}011_2$ (5)

4.4 $1101111{,}1101111_2 \div 101{,}011_2$ (5)

[20]

QUESTION 5

Draw the circuit diagrams of the Boolean expressions below. Do not change or simplify the expressions:

5.1 $$\overline{(A \oplus \overline{B \bullet C}) \oplus C \bullet (\overline{A} \bullet \overline{B + \overline{C}})} = f$$ (6)

5.2 $$\overline{(A + \overline{C \bullet B}) \oplus \overline{(C \bullet \overline{A} + B)}} = f$$ (5)

5.3 $$\overline{\overline{A} \oplus B \oplus C \oplus D \oplus E} = f$$ (4)

[15]

QUESTION 6

Design a circuit with THREE inputs making use of TWO exclusive OR gates and a NOT gate.

The circuit must supply a binary ONE at the output if there is an UNEVEN number of binary ZEROES on the input, and if there is an EVEN number of ZEROES on the input the output must be a binary ZERO.

The following must be shown clearly on your design:

6.1 Truth Table (2)

6.2 Boolean Manipulation (6)

6.3 Circuit diagram (2)

[10]

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QUESTION 7

7.1 Make a neatly labelled circuit diagram of a THREE-BIT asynchronous binary DOWN counter. Indicate the most and least significant bits on the circuit diagram.

Give the timing diagram for TEN clock pulses. (10)

7.2 Make a neat sketch of a RS master/slave flip-flop which makes use of only NAND gates. Show all input and output points. (8)

7.3 Discuss the purpose of the set and reset lines on a flip-flop. (4)

7.4 Explain why a master-slave flip-flop clocks on the trailing edge. (8)
[30]

TOTAL SECTION B: 75
GRAND TOTAL: 100

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